

SAT Unit 1

CB Bank: Step-by-Step Solutions

Novel Prep

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86 worked solutions across all five original sections.

Each problem retains its original question, answer choices, tables, and graphs. The solution follows in separate numbered steps, with a highlighted final answer. Original question numbering restarts in each section; labels such as 1.3 / Q8 identify the section and question unambiguously.

Long questions or solutions continue onto the following page at a step boundary so that text and graphs remain readable.

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Source notes

The supplied question wording and graphics are reproduced without alteration. In 1.1 / Q5, option D has an unmatched closing parenthesis in the source; the correct answer is A. In 1.3 / Q18, the horizontal tick spacing is uneven; both labeled intercepts consistently give a total of \$160. In 1.4 / Q20, only one line is distinguishable in the supplied graph; the added line is parallel and distinct from that visible line, which is sufficient to rule out any common solution.

1.1 Linear Equations in One Variable

1.1 / Question 1

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

A. 3

B. 7

C. 8

D. 12

Worked solution: 1.1 / Q1

Step 1. Find the removal rate.

In 5 hours, the auger removes $24,000 - 19,350 = 4,650$ bushels. Thus the rate is $4,650 \div 5 = 930$ bushels per hour.

Step 2. Find the total amount removed.

To leave 12,840 bushels, Hector must remove $24,000 - 12,840 = 11,160$ bushels.

Step 3. Find the total elapsed time.

$t = 11,160 \div 930 = 12$ hours. Check: $24,000 - 930(12) = 12,840$. This is the total time, including the first 5 hours.

Final answer: D; 12 hours

1.1 / Question 2

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by \$5. Assuming gasoline costs \$4 per gallon, which equation can Alan use to determine how many fewer average miles, m , he should drive each week?

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

Worked solution: 1.1 / Q2

Step 1. Convert fuel efficiency to cost per mile.

Each mile uses $\frac{1}{25}$ gallon, so the cost per mile is $4\left(\frac{1}{25}\right) = \frac{4}{25}$ dollar.

Step 2. Write the savings expression.

Driving m fewer miles saves $\frac{4}{25}m$ dollars each week.

Step 3. Set the savings equal to the goal.

$\frac{4}{25}m = 5$. This gives $m = 31.25$ fewer miles; the equation requested is choice D.

Final answer: D; $\frac{4}{25}m = 5$

1.1 / Question 3

How many solutions does the equation $10(15x - 9) = -15(6 - 10x)$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Worked solution: 1.1 / Q3

Step 1. Expand the left side.

$$10(15x - 9) = 150x - 90.$$

Step 2. Expand the right side.

$$-15(6 - 10x) = -90 + 150x.$$

Step 3. Compare the expressions.

Both sides are identical for every real x . Subtracting $150x$ gives the true statement $-90 = -90$.

Final answer: C; infinitely many solutions

1.1 / Question 4

In the given equation, k is a constant. The equation has no solution. What is the value of k ?

$$3(kx + 13) = \frac{48}{17}x + 36$$

Worked solution: 1.1 / Q4

Step 1. Expand and collect the variable terms.

$$3kx + 39 = \frac{48}{17}x + 36, \text{ so } \left(3k - \frac{48}{17}\right)x = -3.$$

Step 2. Apply the no-solution condition.

The coefficient of x must be zero, leaving the false statement $0 = -3$. Thus $3k = \frac{48}{17}$.

Step 3. Solve and check.

$k = \frac{16}{17}$. Substitution gives $\frac{48}{17}x + 39 = \frac{48}{17}x + 36$, which is impossible.

Final answer: $\frac{16}{17}$

1.1 / Question 5

A certain product costs a company \$65 to make. The product is sold by a salesperson who earns a commission that is equal to 20% of the sales price of the product. The profit the company makes for each unit is equal to the sales price minus the combined cost of making the product and the commission. If the sales price of the product is \$100, which of the following equations gives the number of units, u , of the product the company sold to make a profit of \$6,840?

A. $(100(1 - 0.2) - 65)u = 6840$

B. $(100 - 65)(1 - 0.8)u = 6840$

C. $0.8(100) - 65u = 6840$

D. $0.2(100) + 65)u = 6840$

Worked solution: 1.1 / Q5

Step 1. Find the commission per unit.

The commission is $0.20(100) = 20$ dollars.

Step 2. Find the profit per unit.

Profit is sales price minus manufacturing cost minus commission: $100 - 65 - 20 = 15$ dollars.

Step 3. Model the total profit.

For u units, $(100(1 - 0.2) - 65)u = 6,840$. Check: $u = 6,840/15 = 456$ units.

Final answer: A; $(100(1 - 0.2) - 65)u = 6840$

1.1 / Question 6

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k ?

$$2(kx - n) = -\frac{28}{15}x - \frac{36}{19}$$

Worked solution: 1.1 / Q6

Step 1. Expand the equation.

$$2kx - 2n = -\frac{28}{15}x - \frac{36}{19}.$$

Step 2. Make the coefficients of x equal.

For no solution, $2k = -\frac{28}{15}$, so $k = -\frac{14}{15}$.

Step 3. Check that the constants differ.

Equal constants would require $-2n = -\frac{36}{19}$, or $n = \frac{18}{19}$. But $n > 1$, so the constants cannot be equal. The equation therefore has no solution.

Final answer: $-\frac{14}{15}$

1.1 / Question 7

What value of t is the solution to the given equation?

$$5(t + 3) - 7(t + 3) = 38$$

Worked solution: 1.1 / Q7

Step 1. Combine the common factor.

$$5(t + 3) - 7(t + 3) = (5 - 7)(t + 3) = -2(t + 3).$$

Step 2. Isolate t .

$$-2(t + 3) = 38 \text{ gives } t + 3 = -19, \text{ so } t = -22.$$

Step 3. Check in the original equation.

$$5(-19) - 7(-19) = -95 + 133 = 38.$$

Final answer: -22

1.1 / Question 8

Townsend Realty Group Investments

Property address	Purchase price (dollars)	Monthly rental price (dollars)
Clearwater Lane	128,000	950
Driftwood Drive	176,000	1,310
Edgemont Street	70,000	515
Glenview Street	140,000	1,040
Hamilton Circle	450,000	3,365

The Townsend Realty Group invested in the five different properties listed in the table above. The table shows the amount, in dollars, the company paid for each property and the corresponding monthly rental price, in dollars, the company charges for the property at each of the five locations.

Townsend Realty purchased the Glenview Street property and received a 40% discount off the original price along with an additional 20% off the discounted price for purchasing the property in cash. Which of the following best approximates the original price, in dollars, of the Glenview Street property?

- A. \$350,000
- B. \$291,700
- C. \$233,300
- D. \$175,000

Worked solution: 1.1 / Q8**Step 1. Represent the first discount.**

Let P be the original price. A 40% discount leaves $0.60P$.

Step 2. Apply the second discount.

An additional 20% off leaves $0.80(0.60P) = 0.48P$. The table gives the paid price as \$140,000.

Step 3. Solve for the original price.

$0.48P = 140,000$, so $P = 291,666.\bar{6} \approx 291,700$. Successive discounts are multiplied, not added.

Final answer: B; approximately \$291,700

1.1 / Question 9

Each side of a 30-sided polygon has one of three lengths. The number of sides with length 8 **centimeters (cm)** is 5 times the number of sides n with length 3 cm. There are 6 sides with length 4 cm. Which equation must be true for the value of n ?

A. $5n + 6 = 30$

B. $6n + 6 = 30$

C. $8n + 3n + 4n = 30$

D. $8(5n) + 3n + 4(6) = 30$

Worked solution: 1.1 / Q9

Step 1. Count the sides of each length.

There are n sides of length 3 cm, $5n$ sides of length 8 cm, and 6 sides of length 4 cm.

Step 2. Use the total number of sides.

$n + 5n + 6 = 30$, so $6n + 6 = 30$.

Step 3. Check the interpretation.

The equation gives $n = 4$: there are $4 + 20 + 6 = 30$ sides. The problem counts sides, rather than adding their lengths.

Final answer: B; $6n + 6 = 30$

1.1 / Question 10

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

$$-3x + 21px = 84$$

A. $p = 0$

B. $p = \frac{1}{7}$

C. $p = \frac{4}{3}$

D. $p = 4$

Worked solution: 1.1 / Q10

Step 1. Factor out x .

$$-3x + 21px = (-3 + 21p)x, \text{ so } (-3 + 21p)x = 84.$$

Step 2. Force an impossible equation.

If the coefficient were nonzero, one solution would exist. For no solution, set $-3 + 21p = 0$.

Step 3. Solve and verify.

$21p = 3$, so $p = \frac{1}{7}$. The original equation becomes $0 = 84$, which is false.

Final answer: B; $p = \frac{1}{7}$

1.1 / Question 11

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x+6$ is between which of the following pairs of values?

A. -7 and -3

B. -2 and 2

C. 2 and 7

D. 8 and 13

Worked solution: 1.1 / Q11

Step 1. Clear the denominators.

Multiply $\frac{x+6}{3} = \frac{x+6}{13}$ by 39 to obtain $13(x+6) = 3(x+6)$.

Step 2. Solve for the requested expression.

Subtract $3(x+6)$: $10(x+6) = 0$. Hence $x+6 = 0$.

Step 3. Locate the value.

Since $-2 < 0 < 2$, the requested expression is between -2 and 2 . The question asks for $x+6$, not x .

Final answer: B; between -2 and 2

1.1 / Question 12

The equation $9x + 5 = a(x + b)$, where a and b are constants, has no solutions. Which of the following must be true?

- I. $a = 9$
- II. $b = 5$
- III. $b \neq \frac{5}{9}$

A. None

B. I only

C. I and II only

D. I and III only

Worked solution: 1.1 / Q12

Step 1. Expand the right side.

$$9x + 5 = ax + ab.$$

Step 2. Match coefficients but not constants.

No solution requires $a = 9$ and $ab \neq 5$. If the coefficients differed, there would be one solution.

Step 3. Evaluate the statements.

With $a = 9$, the constant condition is $9b \neq 5$, or $b \neq \frac{5}{9}$. Thus I and III must hold; II need not hold.

Final answer: D; I and III only

1.1 / Question 13

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

A. $5x$

B. $10x$

C. $2x + 20$

D. $10x + 20$

Worked solution: 1.1 / Q13

Step 1. Count the stations.

Experiment A uses x stations, so Experiment B uses $5 - x$ stations.

Step 2. Write the salt requirement.

A needs $6x$ teaspoons; B needs $4(5 - x)$ teaspoons. Total: $6x + 4(5 - x)$.

Step 3. Simplify and check.

$6x + 20 - 4x = 2x + 20$. At $x = 0$, all five stations use B and need 20 teaspoons, consistent with the formula.

Final answer: C; $2x + 20$

1.2 Linear Functions

1.2 / Question 1

A window repair specialist charges **\$220** for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is **\$400**. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

Worked solution: 1.2 / Q1

Step 1. Find the additional hourly rate.

The extra $5 - 2 = 3$ hours cost $400 - 220 = 180$ dollars. Rate: $180/3 = 60$ dollars per hour.

Step 2. Build the cost function.

For $x \geq 2$, there are $x - 2$ additional hours, so $f(x) = 220 + 60(x - 2)$.

Step 3. Simplify and check.

$f(x) = 60x + 100$. Check: $f(2) = 220$ and $f(5) = 400$.

Final answer: A; $f(x) = 60x + 100$

1.2 / Question 2

An economist modeled the demand Q for a certain product as a linear function of the selling price P . The demand was 20,000 units when the selling price was \$40 per unit, and the demand was 15,000 units when the selling price was \$60 per unit. Based on the model, what is the demand, in units, when the selling price is \$55 per unit?

A. 16,250

B. 16,500

C. 16,750

D. 17,500

Worked solution: 1.2 / Q2

Step 1. Find the slope of demand.

$$m = \frac{15,000 - 20,000}{60 - 40} = -250 \text{ units per dollar.}$$

Step 2. Find the price increase.

The price rises from \$40 to \$55, an increase of \$15. Demand therefore changes by $-250(15) = -3,750$ units.

Step 3. Calculate the new demand.

$$Q = 20,000 - 3,750 = 16,250 \text{ units.}$$

Final answer: A; 16,250 units

1.2 / Question 3

The cost of renting a backhoe for up to 10 days is \$270 for the first day and \$135 for each additional day. Which of the following equations gives the cost y , in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$

Worked solution: 1.2 / Q3

Step 1. Count the additional days.

The first day costs \$270; the other $x - 1$ days each cost \$135.

Step 2. Write the total cost.

$y = 270 + 135(x - 1)$, for integers $1 \leq x \leq 10$.

Step 3. Simplify and check.

$y = 135x + 135$. At $x = 1$, the cost is \$270, as required.

Final answer: D; $y = 135x + 135$

1.2 / Question 4

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

A. 16.38

B. 48.38

C. 475.29

D. 507.29

Worked solution: 1.2 / Q4

Step 1. Identify the rate of temperature change.

The slope of $F(x) = \frac{9}{5}(x - 273.15) + 32$ is $\frac{9}{5}$ Fahrenheit degrees per kelvin.

Step 2. Apply the slope to the increase.

$$\Delta F = \frac{9}{5}\Delta x = \frac{9}{5}(9.10).$$

Step 3. Calculate the increase.

$\Delta F = 16.38$ degrees Fahrenheit. The constant offsets cancel when finding a change.

Final answer: A; 16.38 degrees Fahrenheit

1.2 / Question 5

The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x -coordinate of the x -intercept of the graph of $y = h(x)$ in the xy -plane?

Worked solution: 1.2 / Q5

Step 1. Add the functions.

$$h(x) = \left(\frac{1}{4}x - 9\right) + \left(\frac{3}{4}x + 21\right) = x + 12.$$

Step 2. Use the definition of an x -intercept.

At an x -intercept, the output is zero: $h(x) = 0$.

Step 3. Solve and check.

$x + 12 = 0$, so $x = -12$. Check: $f(-12) = -12$ and $g(-12) = 12$, giving $h(-12) = 0$.

Final answer: -12

1.2 / Question 6

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function F defined below, where $F(x)$ is the average price per gallon x months after September 1.

$$F(x) = 2.74 - 0.19(x - 3)$$

The constant 2.74 in this function estimates which of the following?

- A. The average monthly decrease in the price per gallon
- B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014
- C. The average price per gallon on September 1, 2014
- D. The average price per gallon on December 1, 2014

Worked solution: 1.2 / Q6

Step 1. Identify when the variable term vanishes.

In $F(x) = 2.74 - 0.19(x - 3)$, the term $-0.19(x - 3)$ is zero when $x = 3$.

Step 2. Evaluate the model.

$F(3) = 2.74$, so the constant is the price per gallon 3 months after September 1.

Step 3. Translate the time to a date.

Three months after September 1, 2014, is December 1, 2014.

Final answer: D; the average price per gallon on December 1, 2014

1.2 / Question 7

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Function f is defined by $f(x) = ax + b$, where a and b are constants. What is the value of $a - b$?

x	$f(x)$
1	-64
2	0
3	64

A. -64

B. 62

C. 128

D. 192

Worked solution: 1.2 / Q7

Step 1. Find a , the slope.

$a = \frac{0 - (-64)}{2 - 1} = 64$. The next interval also has slope 64.

Step 2. Find the intercept b .

Using $(2, 0)$ in $f(x) = 64x + b$ gives $0 = 128 + b$, so $b = -128$.

Step 3. Compute the requested difference.

$a - b = 64 - (-128) = 192$.

Final answer: D; 192

1.2 / Question 8

Oil and gas production in a certain area dropped from 4 million barrels in 2000 to 1.9 million barrels in 2013. Assuming that the oil and gas production decreased at a constant rate, which of the following linear functions f best models the production, in millions of barrels, t years after the year 2000?

A. $f(t) = \frac{21}{130}t + 4$

B. $f(t) = \frac{19}{130}t + 4$

C. $f(t) = -\frac{21}{130}t + 4$

D. $f(t) = -\frac{19}{130}t + 4$

Worked solution: 1.2 / Q8

Step 1. Write the two time-production points.

The data give $(0, 4)$ and $(13, 1.9)$, with production measured in millions of barrels.

Step 2. Find the slope.

$$m = \frac{1.9-4}{13} = -\frac{2.1}{13} = -\frac{21}{130}.$$

Step 3. Write and verify the model.

$$f(t) = -\frac{21}{130}t + 4. \text{ At } t = 13, f(13) = 4 - 2.1 = 1.9.$$

Final answer: C; $f(t) = -\frac{21}{130}t + 4$

1.2 / Question 9

One gallon of paint will cover 220 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P , in gallons, needed to paint the walls of the room twice?

A. $P = \frac{w}{110}$

B. $P = 440w$

C. $P = \frac{w}{220}$

D. $P = 220w$

Worked solution: 1.2 / Q9

Step 1. Account for two coats.

Painting w square feet twice requires coverage of $2w$ square feet.

Step 2. Divide by coverage per gallon.

$$P = \frac{2w}{220} \text{ gallons.}$$

Step 3. Simplify and check the units.

$$P = \frac{w}{110}. \text{ Square feet divided by square feet per gallon gives gallons.}$$

Final answer: A; $P = \frac{w}{110}$

1.2 / Question 10

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

A. 3.78

B. 35.78

C. 487.89

D. 519.89

Worked solution: 1.2 / Q10

Step 1. Identify the slope.

Each increase of 1 kelvin increases the Fahrenheit temperature by $\frac{9}{5}$ degrees.

Step 2. Multiply the temperature change.

$$\Delta F = \frac{9}{5}(2.10).$$

Step 3. Calculate the result.

$\Delta F = 3.78$ degrees Fahrenheit. Do not add 32 when converting a temperature difference.

Final answer: A; 3.78 degrees Fahrenheit

1.2 / Question 11

The table above shows some values of x and their corresponding values $f(x)$ for the linear function f . What is the x -intercept of the graph of $y = f(x)$ in the xy -plane?

x	-11	-10	-9	-8
$f(x)$	21	18	15	12

A. $(-3, 0)$

B. $(-4, 0)$

C. $(-9, 0)$

D. $(-12, 0)$

Worked solution: 1.2 / Q11

Step 1. Find the slope from the table.

When x increases by 1, $f(x)$ decreases by 3. Thus $m = -3$.

Step 2. Find the rule.

Using $(-8, 12)$, $12 = -3(-8) + b$ gives $b = -12$. Hence $f(x) = -3x - 12$.

Step 3. Set the output equal to zero.

$-3x - 12 = 0$ gives $x = -4$. The x -intercept is the point $(-4, 0)$.

Final answer: B; $(-4, 0)$

1.2 / Question 12

Macronutrient	Food calories	Kilojoules
Protein	4.0	16.7
Fat	9.0	37.7
Carbohydrate	4.0	16.7

The table above gives the typical amounts of energy per gram, expressed in both food calories and kilojoules, of the three macronutrients in food. If the 180 food calories in a granola bar come entirely from p grams of protein, f grams of fat, and c grams of carbohydrate, which of the following expresses f in terms of p and c ?

A. $f = 20 + \frac{4}{9}(p + c)$

B. $f = 20 - \frac{4}{9}(p + c)$

C. $f = 20 - \frac{4}{9}p - c$

D. $f = 20 + \frac{9}{4}(p + c)$

Worked solution: 1.2 / Q12

Step 1. Use food calories per gram.

Protein and carbohydrate each contribute 4 calories per gram; fat contributes 9. Thus $4p + 9f + 4c = 180$.

Step 2. Isolate the fat term.

$$9f = 180 - 4p - 4c.$$

Step 3. Divide by 9 and factor.

$$f = 20 - \frac{4}{9}p - \frac{4}{9}c = 20 - \frac{4}{9}(p + c). \text{ Use the calorie column, since 180 is measured in food calories.}$$

Final answer: B; $f = 20 - \frac{4}{9}(p + c)$

1.2 / Question 13

An object hangs from a spring. The formula $\ell = 30 + 2w$ relates the length ℓ , in centimeters, of the spring to the weight w , in newtons, of the object. Which of the following describes the meaning of the 2 in this context?

- A. The length, in centimeters, of the spring with no weight attached
- B. The weight, in newtons, of an object that will stretch the spring 30 centimeters
- C. The increase in the weight, in newtons, of the object for each one-centimeter increase in the length of the spring
- D. The increase in the length, in centimeters, of the spring for each one-newton increase in the weight of the object

Worked solution: 1.2 / Q13

Step 1. Identify input and output.

In $\ell = 30 + 2w$, w is weight in newtons and ℓ is length in centimeters.

Step 2. Interpret the coefficient.

The slope is $\frac{\Delta\ell}{\Delta w} = 2$ centimeters per newton.

Step 3. State the meaning.

Each additional 1 newton of weight increases the spring's length by 2 centimeters. The 30 represents the length when no weight is attached.

Final answer: D; length increases by 2 cm for each additional newton

1.2 / Question 14

For the function f , if $f(3x) = x - 6$ for all values of x , what is the value of $f(6)$?

A. -6

B. -4

C. 0

D. 2

Worked solution: 1.2 / Q14

Step 1. Match the input to 6.

The input of f is $3x$. To evaluate $f(6)$, solve $3x = 6$, giving $x = 2$.

Step 2. Substitute into the given identity.

$$f(3 \cdot 2) = 2 - 6.$$

Step 3. Evaluate.

$f(6) = -4$. Substituting $x = 6$ instead would calculate $f(18)$.

Final answer: B; -4

1.2 / Question 15

For groups of 25 or more people, a museum charges \$21 per person for the first 25 people and \$14 for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

A. $f(n) = 14n + 175$

B. $f(n) = 14n + 525$

C. $f(n) = 35n - 350$

D. $f(n) = 14n + 21$

Worked solution: 1.2 / Q15

Step 1. Find the cost for the first 25 people.

$$25(21) = 525 \text{ dollars.}$$

Step 2. Charge for the remaining people.

There are $n - 25$ additional people, costing $14(n - 25)$ dollars.

Step 3. Combine and simplify.

$$f(n) = 525 + 14(n - 25) = 14n + 175, \text{ for } n \geq 25. \text{ Check: } f(25) = 525.$$

Final answer: A; $f(n) = 14n + 175$

1.3 Linear Equations in Two Variables

1.3 / Question 1

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

x	y
k	13
$k + 7$	-15

Worked solution: 1.3 / Q1

Step 1. Use the y -intercept's x -coordinate.

Every y -intercept has $x = 0$, so $k - 5 = 0$ and $k = 5$. The points are $(5, 13)$ and $(12, -15)$.

Step 2. Find the slope.

$$m = \frac{-15-13}{12-5} = -4.$$

Step 3. Find b .

In $y = -4x + b$, use $(5, 13)$: $13 = -20 + b$, so $b = 33$. Check: $-4(12) + 33 = -15$.

Final answer: 33

1.3 / Question 2

The graph of the equation $ax + ky = 6$ is a line in the xy -plane, where a and k are constants. If the line contains the points $(-2, -6)$ and $(0, -3)$, what is the value of k ?

A. -2

B. -1

C. 2

D. 3

Worked solution: 1.3 / Q2

Step 1. Choose the point that eliminates a .

The point $(0, -3)$ lies on $ax + ky = 6$.

Step 2. Substitute its coordinates.

$a(0) + k(-3) = 6$, so $-3k = 6$.

Step 3. Solve and check consistency.

$k = -2$. The other point gives $-2a + 12 = 6$, so $a = 3$, confirming a consistent line.

Final answer: A; -2

1.3 / Question 3

For line h , the table shows three values of x and their corresponding values of y . Line k is the result of translating line h down 5 units in the xy -plane. What is the x -intercept of line k ?

x	y
18	130
23	160
26	178

A. $\left(-\frac{26}{3}, 0\right)$

B. $\left(-\frac{9}{2}, 0\right)$

C. $\left(-\frac{11}{3}, 0\right)$

D. $\left(-\frac{17}{6}, 0\right)$

Worked solution: 1.3 / Q3

Step 1. Find the equation of line h .

$m = (160 - 130)/(23 - 18) = 6$. Using $(18, 130)$ gives $130 = 6(18) + b$, so $b = 22$ and $h : y = 6x + 22$.

Step 2. Translate down 5 units.

Subtract 5 from every output: line k is $y = 6x + 17$.

Step 3. Find its x -intercept.

Set $y = 0$: $6x + 17 = 0$, giving $x = -\frac{17}{6}$.

Final answer: D; $\left(-\frac{17}{6}, 0\right)$

1.3 / Question 4

A certain apprentice has enrolled in 85 hours of training courses. The equation $10x + 15y = 85$ represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

Worked solution: 1.3 / Q4

Step 1. Interpret the on-site coefficient.

In $10x + 15y = 85$, the term $10x$ means each on-site course requires 10 hours.

Step 2. Interpret the online coefficient.

The term $15y$ means each online course requires 15 hours.

Step 3. Subtract the course lengths.

$15 - 10 = 5$ more hours per online course.

Final answer: 5 hours

1.3 / Question 5

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

Worked solution: 1.3 / Q5

Step 1. Put line p in slope-intercept form.

$4y + 8x = 6$ gives $4y = -8x + 6$, hence $y = -2x + \frac{3}{2}$. Its slope is -2 .

Step 2. Use perpendicular slopes.

For these nonvertical lines, the slopes multiply to -1 : $(-2)m_r = -1$.

Step 3. Solve and check.

$m_r = \frac{1}{2}$, and $(-2)(\frac{1}{2}) = -1$.

Final answer: $\frac{1}{2}$

1.3 / Question 6

In the xy -plane, line k intersects the y -axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k , what is the value of w ?

Worked solution: 1.3 / Q6

Step 1. Find the slope.

Using $(0, -6)$ and $(2, 2)$, $m = \frac{2 - (-6)}{2 - 0} = 4$.

Step 2. Write the equation.

The y -intercept is -6 , so $y = 4x - 6$.

Step 3. Evaluate at $x = 20$.

$w = 4(20) - 6 = 74$. Check: the rule also gives $y = 2$ at $x = 2$.

Final answer: 74

1.3 / Question 7

In the xy -plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 2$. If line ℓ also passes through the point $(3, d)$, what is the value of d ?

Worked solution: 1.3 / Q7

Step 1. Use the parallel-line condition.

The line $y = 8x + 2$ has slope 8, so line ℓ also has slope 8.

Step 2. Use the point $(0, 0)$.

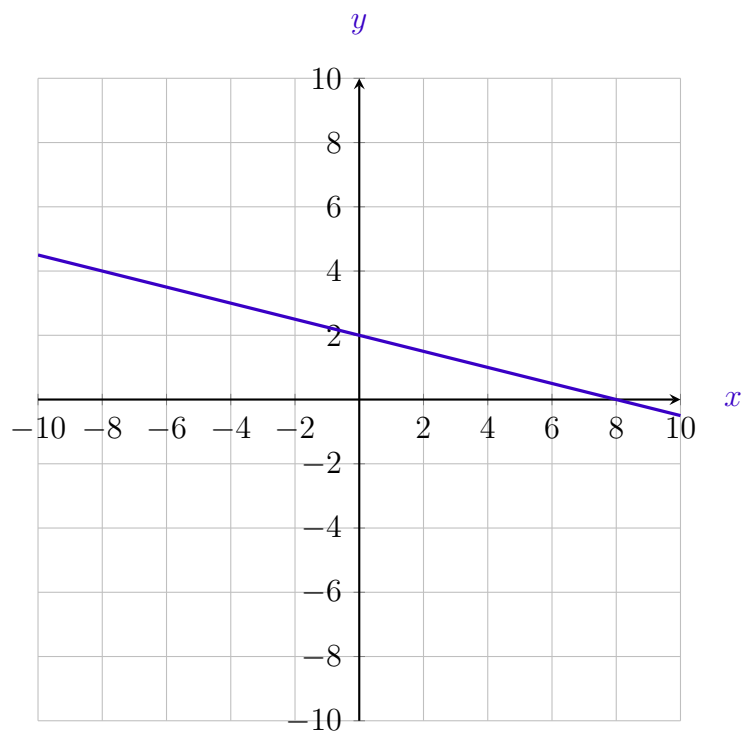
The y -intercept is 0, making the equation $y = 8x$.

Step 3. Evaluate at $x = 3$.

$$d = 8(3) = 24.$$

Final answer: 24

1.3 / Question 8



The graph of $y = f(x) + 14$ is shown. Which equation defines function f ?

A. $f(x) = -\frac{1}{4}x - 12$

B. $f(x) = -\frac{1}{4}x + 16$

C. $f(x) = -\frac{1}{4}x + 2$

D. $f(x) = -\frac{1}{4}x - 14$

Worked solution: 1.3 / Q8**Step 1. Read two intercepts from the graph.**

The displayed line passes through $(0, 2)$ and $(8, 0)$. Its slope is $(0 - 2)/(8 - 0) = -\frac{1}{4}$.

Step 2. Write the displayed equation.

The graph represents $y = f(x) + 14$, so $f(x) + 14 = -\frac{1}{4}x + 2$.

Step 3. Undo the vertical shift.

Subtract 14: $f(x) = -\frac{1}{4}x - 12$. Check: $f(0) + 14 = -12 + 14 = 2$, matching the graph.

Final answer: A; $f(x) = -\frac{1}{4}x - 12$

1.3 / Question 9

Keenan made **32** cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

Worked solution: 1.3 / Q9

Step 1. Identify what y counts.

The variable y is the number of large jars.

Step 2. Interpret the coefficient 5.

Each large jar contains 5 cups of broth, so $5y$ is cups per jar times the number of jars.

Step 3. State the resulting quantity.

The product $5y$ is the total number of cups in all the large jars. Along with $3x$, it adds to 32 cups.

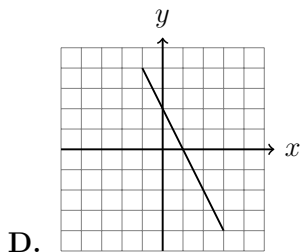
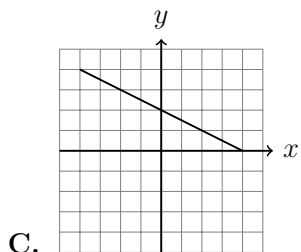
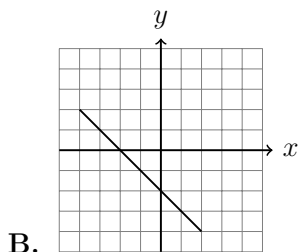
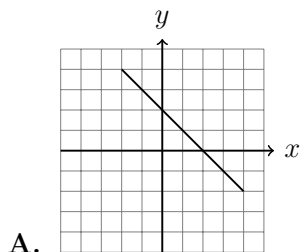
Final answer: C; the total cups of broth in the large jars

1.3 / Question 10

$$ax + by = b$$

In the equation above, a and b are constants and $0 < a < b$. Which of the following could represent the graph of the equation in the xy -plane?

For each length of the square, it counts 0.5



Worked solution: 1.3 / Q10**Step 1. Rewrite the equation.**

Since $b > 0$, divide $ax + by = b$ by b : $y = -\frac{a}{b}x + 1$.

Step 2. Use the condition $0 < a < b$.

Then $0 < \frac{a}{b} < 1$, so the slope is between -1 and 0 . The y -intercept is 1 , and the x -intercept b/a is greater than 1 .

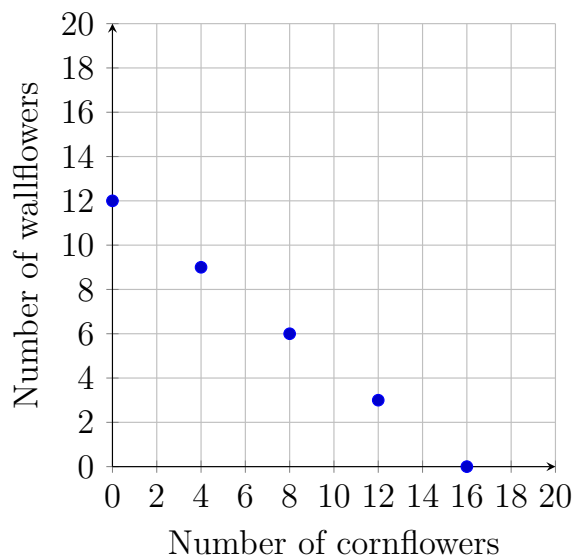
Step 3. Compare the graphs using the scale.

Each square represents 0.5 unit. Choice C crosses the y -axis at 1 and has a negative slope less steep than -1 . A has slope -1 ; B has a negative y -intercept; D is too steep.

Final answer: C

1.3 / Question 11

The points plotted in the coordinate plane above represent the possible numbers of wallflowers and cornflowers that someone can buy at the Garden Store in order to spend exactly \$24.00 total on the two types of flowers. The price of each wallflower is the same and the price of each cornflower is the same. What is the price, in dollars, of 1 cornflower?



Worked solution: 1.3 / Q11

Step 1. Read the all-cornflower point.

The graph includes $(16, 0)$: 16 cornflowers and no wallflowers cost \$24.

Step 2. Write a one-variable equation.

Let c be the price of one cornflower. Then $16c = 24$.

Step 3. Solve and check.

$c = 24/16 = 1.50$. As a check, $(0, 12)$ gives \$2 per wallflower; $(8, 6)$ then costs $8(1.50) + 6(2) = 24$ dollars.

Final answer: \$1.50

1.3 / Question 12

The graph of $7x + 2y = -31$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $\frac{b}{a}$?

A. $-\frac{7}{2}$

B. $-\frac{2}{7}$

C. $\frac{2}{7}$

D. $\frac{7}{2}$

Worked solution: 1.3 / Q12

Step 1. Find the x -intercept.

Set $y = 0$: $7a = -31$, so $a = -\frac{31}{7}$.

Step 2. Find the y -intercept.

Set $x = 0$: $2b = -31$, so $b = -\frac{31}{2}$.

Step 3. Compute the ratio.

$\frac{b}{a} = \frac{-31/2}{-31/7} = \frac{7}{2}$. Both intercepts are negative, so the ratio is positive.

Final answer: D; $\frac{7}{2}$

1.3 / Question 13

Line ℓ in the xy -plane is perpendicular to the line with equation $x = 2$. What is the slope of line ℓ ?

A. 0

B. $\frac{1}{2}$

C. -2

D. The slope of line ℓ is undefined.

Worked solution: 1.3 / Q13

Step 1. Identify the given line.

The equation $x = 2$ describes a vertical line.

Step 2. Determine the perpendicular direction.

A line perpendicular to a vertical line is horizontal, so its equation has the form $y = c$.

Step 3. Find its slope.

A horizontal line has no vertical change: slope = 0.

Final answer: A; 0

1.3 / Question 14

The graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. What is the x -coordinate of the x -intercept of the resulting graph?

Worked solution: 1.3 / Q14

Step 1. Solve the original equation for y .

$$9x - 10y = 19 \text{ gives } y = \frac{9}{10}x - \frac{19}{10}.$$

Step 2. Translate down 4 units.

$$\text{The new equation is } y = \frac{9}{10}x - \frac{19}{10} - 4 = \frac{9}{10}x - \frac{59}{10}.$$

Step 3. Find the new x -intercept.

Set $y = 0$: $9x - 59 = 0$, so $x = \frac{59}{9}$. At that input, the original graph has height 4, which shifts to 0.

Final answer: $\frac{59}{9}$

1.3 / Question 15

The table above shows the coordinates of three points on a line in the xy -plane, where k and n are constants. If the slope of the line is 2, what is the value of $k + n$?

x	y
3	7
k	11
12	n

Worked solution: 1.3 / Q15

Step 1. Find the line's equation.

Using slope 2 and $(3, 7)$, write $y = 2x + b$. Then $7 = 6 + b$, so $y = 2x + 1$.

Step 2. Find k and n .

At $(k, 11)$, $11 = 2k + 1$ gives $k = 5$. At $(12, n)$, $n = 2(12) + 1 = 25$.

Step 3. Add the values.

$$k + n = 5 + 25 = 30.$$

Final answer: 30

1.3 / Question 16

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$5x + 7y = 1$$

$$ax + by = 1$$

A. $10x + 7y = 1$ and $2ax + 5by = 1$

B. $10x + 7y = 1$ and $ax + 2by = 1$

C. $10x + 7y = 1$ and $2ax + by = 1$

D. $5x - 7y = 1$ and $ax + by = 1$

Worked solution: 1.3 / Q16

Step 1. Find the original slopes.

The first line has slope $-\frac{5}{7}$. Its perpendicular partner must have slope $\frac{7}{5}$, so $-a/b = \frac{7}{5}$.

Step 2. Track the slopes in choice B.

For $10x + 7y = 1$, the slope is $-\frac{10}{7}$. For $ax + 2by = 1$, the slope is $-\frac{a}{2b} = \frac{7}{10}$.

Step 3. Check perpendicularity.

Their product is $(-\frac{10}{7})(\frac{7}{10}) = -1$. For A, C, and D, the slope products are $-\frac{4}{5}$, -4 , and 1 , respectively.

Final answer: B

1.3 / Question 17

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won G green tokens and R red tokens worth a total of 380 points. The given equation represents this situation. How many more points is a red token worth than a green token?

$$5G + 45R = 380$$

Worked solution: 1.3 / Q17

Step 1. Interpret the green-token coefficient.

In $5G + 45R = 380$, each green token is worth 5 points.

Step 2. Interpret the red-token coefficient.

Each red token is worth 45 points.

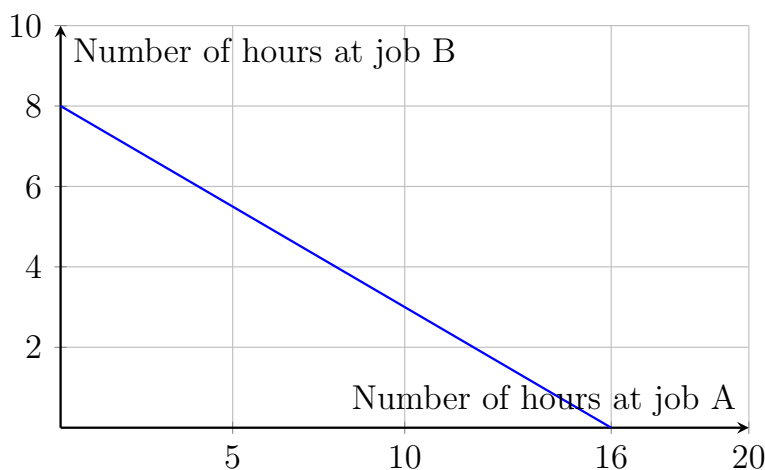
Step 3. Find the difference.

$45 - 5 = 40$ more points per red token.

Final answer: 40 points

1.3 / Question 18

To earn money for college, Avery works two part-time jobs: A and B. She earns \$10 per hour working at job A and \$20 per hour working at job B. In one week, Avery earned a total of s dollars for working at the two part-time jobs. The graph below represents all possible combinations of numbers of hours Avery could have worked at the two jobs to earn s dollars. What is the value of s ?



A. 128

B. 160

C. 200

D. 320

Worked solution: 1.3 / Q18

Step 1. Read an intercept.

The graph contains $(0, 8)$: Avery can earn the total by working 0 hours at A and 8 hours at B.

Step 2. Use job B's hourly pay.

$$s = 10(0) + 20(8) = 160 \text{ dollars.}$$

Step 3. Check the other intercept.

The graph also contains $(16, 0)$, giving $s = 10(16) + 20(0) = 160$ dollars.

Final answer: B; 160

1.3 / Question 19

The line with the equation $\frac{4}{5}x + \frac{1}{3}y = 1$ is graphed in the xy -plane.

What is the x -coordinate of the x -intercept of the line?

Worked solution: 1.3 / Q19

Step 1. Use the x -intercept condition.

At the x -intercept, $y = 0$.

Step 2. Substitute into the equation.

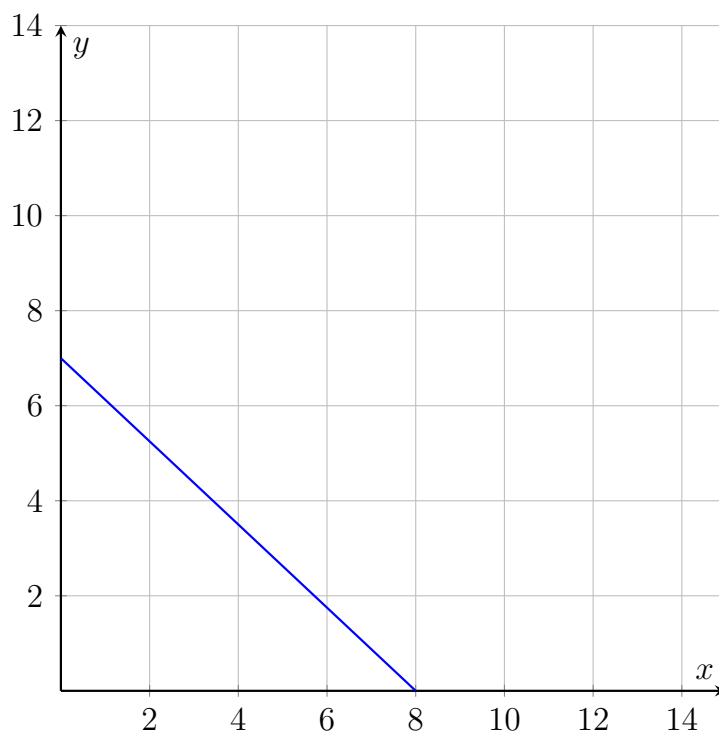
$$\frac{4}{5}x + \frac{1}{3}(0) = 1, \text{ so } \frac{4}{5}x = 1.$$

Step 3. Solve and verify.

$$x = \frac{5}{4}. \text{ Check: } \frac{4}{5} \cdot \frac{5}{4} = 1.$$

Final answer: $\frac{5}{4}$

1.3 / Question 20



The point with coordinates $(d, 4)$ lies on the line shown. What is the value of d ?

A. $\frac{7}{2}$

B. $\frac{26}{7}$

C. $\frac{24}{7}$

D. $\frac{27}{8}$

Worked solution: 1.3 / Q20**Step 1. Read the intercepts.**

The graph passes through $(0, 7)$ and $(8, 0)$, so its slope is $-\frac{7}{8}$.

Step 2. Write the line's equation.

$$y = -\frac{7}{8}x + 7.$$

Step 3. Use the point $(d, 4)$.

$4 = -\frac{7}{8}d + 7$ gives $-3 = -\frac{7}{8}d$, so $d = \frac{24}{7}$. Check: $-\frac{7}{8}(\frac{24}{7}) + 7 = 4$.

Final answer: C; $\frac{24}{7}$

1.3 / Question 21

In the xy -plane, line k is defined by $x + y = 0$. Line j is perpendicular to line k , and the y -intercept of line j is $(0, 3)$. Which of the following is an equation of line j ?

A. $x + y = 3$

B. $x + y = -3$

C. $x - y = 3$

D. $x - y = -3$

Worked solution: 1.3 / Q21

Step 1. Find the slope of line k .

$x + y = 0$ becomes $y = -x$, with slope -1 .

Step 2. Construct the perpendicular line.

Line j has slope 1 and y -intercept 3 , so $y = x + 3$.

Step 3. Match the answer's form.

Rearrange to $x - y = -3$. The slopes satisfy $(-1)(1) = -1$.

Final answer: D; $x - y = -3$

1.3 / Question 22

How many liters of a 25% saline solution must be added to 3 liters of a 10% saline solution to obtain a 15% saline solution?

Worked solution: 1.3 / Q22

Step 1. Define the amount added.

Let v be the liters of 25% solution. Initial salt volume is $0.10(3) = 0.30$ liter; added salt is $0.25v$ liter.

Step 2. Conserve the amount of salt.

The final mixture has $v + 3$ liters at 15% concentration: $0.30 + 0.25v = 0.15(v + 3)$.

Step 3. Solve and check.

$0.10v = 0.15$, so $v = 1.5$ liters. Check: $(0.30 + 0.375)/4.5 = 0.15$.

Final answer: 1.5 liters

1.4 Systems of Two Linear Equations

1.4 / Question 1

$$-x + y = -3.5$$

$$x + 3y = 9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

Worked solution: 1.4 / Q1

Step 1. Add the two equations.

$(-x + y) + (x + 3y) = -3.5 + 9.5$, so $4y = 6$.

Step 2. Solve for the requested variable.

$y = 6/4 = 1.5$.

Step 3. Check both equations.

The first equation gives $x = 5$. Then $-5 + 1.5 = -3.5$ and $5 + 3(1.5) = 9.5$.

Final answer: 1.5

1.4 / Question 2

$$\begin{aligned}\frac{3}{2}y - \frac{1}{4}x &= \frac{2}{3} - \frac{3}{2}y \\ \frac{1}{2}x + \frac{3}{2} &= py + \frac{9}{2}\end{aligned}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

Worked solution: 1.4 / Q2

Step 1. Simplify the first equation.

$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$ gives $3y - \frac{1}{4}x = \frac{2}{3}$, hence $y = \frac{1}{12}x + \frac{2}{9}$.

Step 2. Simplify the second equation.

$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$ gives $py = \frac{1}{2}x - 3$. For $p \neq 0$, its slope is $\frac{1}{2p}$. If $p = 0$, this is a vertical line and intersects the first line once.

Step 3. Match slopes and check intercepts.

$\frac{1}{2p} = \frac{1}{12}$ gives $p = 6$. The second line becomes $y = \frac{1}{12}x - \frac{1}{2}$, whose intercept differs from $\frac{2}{9}$, so the lines never intersect.

Final answer: 6

1.4 / Question 3

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

Worked solution: 1.4 / Q3

Step 1. Simplify the first equation.

$4x - 6y = 10y + 2$ gives $4x - 16y = 2$, or $y = \frac{1}{4}x - \frac{1}{8}$.

Step 2. Write the second line's slope.

From $ty = \frac{1}{2} + 2x$, the slope is $2/t$ for $t \neq 0$. At $t = 0$, the vertical line intersects the first line, so it cannot give no solution.

Step 3. Match slopes and check constants.

$2/t = 1/4$ gives $t = 8$. The second equation is then $y = \frac{1}{4}x + \frac{1}{16}$, with a different intercept.

Final answer: 8

1.4 / Question 4

During a month, Morgan ran r miles at 5 miles per hour and biked b miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

A. 80

B. 100

C. 120

D. 160

Worked solution: 1.4 / Q4

Step 1. Translate distance into time.

Running time is $r/5$ hours and biking time is $b/10$ hours. Twice as much biking time means $b/10 = 2(r/5)$, so $b = 4r$.

Step 2. Use the total distance.

$r + b = 200$ becomes $r + 4r = 200$. Thus $r = 40$.

Step 3. Find and check the biking distance.

$b = 160$ miles. Running takes $40/5 = 8$ hours and biking takes $160/10 = 16$ hours, exactly twice as long.

Final answer: D; 160 miles

1.4 / Question 5

In the system of equations below, a and c are constants.

$$\frac{1}{2}x + \frac{1}{3}y = \frac{1}{6}$$

$$ax + y = c$$

If the system of equations has an infinite number of solutions (x, y) , what is the value of a ?

A. $-\frac{1}{2}$

B. 0

C. $\frac{1}{2}$

D. $\frac{3}{2}$

Worked solution: 1.4 / Q5

Step 1. Recognize identical-line equations.

Infinitely many solutions require both equations to describe the same line.

Step 2. Normalize the coefficient of y .

Multiply $\frac{1}{2}x + \frac{1}{3}y = \frac{1}{6}$ by 3: $\frac{3}{2}x + y = \frac{1}{2}$.

Step 3. Compare coefficients.

Matching $ax + y = c$ gives $a = \frac{3}{2}$ and $c = \frac{1}{2}$. The requested value is a .

Final answer: D; $\frac{3}{2}$

1.4 / Question 6

$$y = 4x + 1$$

$$4y = 15x - 8$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

Worked solution: 1.4 / Q6

Step 1. Substitute for y .

Use $y = 4x + 1$ in $4y = 15x - 8$: $4(4x + 1) = 15x - 8$.

Step 2. Solve for both variables.

$16x + 4 = 15x - 8$ gives $x = -12$. Then $y = 4(-12) + 1 = -47$.

Step 3. Find the requested difference.

$x - y = -12 - (-47) = 35$. Check: $4(-47) = -188 = 15(-12) - 8$.

Final answer: 35

1.4 / Question 7

$$7x - 5y = 4$$

$$4x - 8y = 9$$

If (x, y) is the solution to the system of equations above, what is the value of $3x + 3y$?

A. -13

B. -5

C. 5

D. 13

Worked solution: 1.4 / Q7

Step 1. Look for the requested expression.

Subtracting the second equation from the first creates $3x + 3y$ directly.

Step 2. Subtract corresponding sides.

$$(7x - 5y) - (4x - 8y) = 4 - 9.$$

Step 3. Simplify.

$3x + 3y = -5$. Both equations hold at the solution, so their difference holds there too; solving separately for x and y is unnecessary.

Final answer: B; -5

1.4 / Question 8

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

$$\begin{aligned}48x - 72y &= 30y + 24 \\ ry &= \frac{1}{6} - 16x\end{aligned}$$

Worked solution: 1.4 / Q8

Step 1. Simplify the first equation.

$48x - 72y = 30y + 24$ gives $48x - 102y = 24$. Divide by -3 : $-16x + 34y = -8$.

Step 2. Compare with the second equation.

$ry = \frac{1}{6} - 16x$ becomes $16x + ry = \frac{1}{6}$. Multiply the simplified first equation by -1 : $16x - 34y = 8$.

Step 3. Match coefficients and compare constants.

Set $r = -34$. The left sides are now identical, but $8 \neq \frac{1}{6}$, so no point satisfies both.

Final answer: -34

1.4 / Question 9

The solution to the given system of equations is (x, y) . What is the value of y ?

$$24x + y = 48$$

$$6x + y = 72$$

Worked solution: 1.4 / Q9

Step 1. Eliminate y .

Subtract $6x + y = 72$ from $24x + y = 48$: $18x = -24$.

Step 2. Find x and substitute.

$x = -\frac{4}{3}$. In $6x + y = 72$, this gives $-8 + y = 72$, so $y = 80$.

Step 3. Check the first equation.

$$24\left(-\frac{4}{3}\right) + 80 = -32 + 80 = 48.$$

Final answer: 80

1.4 / Question 10

A movie theater sells two types of tickets, adult tickets for \$12 and child tickets for \$8. If the theater sold 30 tickets for a total of \$300, how much, in dollars, was spent on adult tickets? (Disregard the \$ sign when gridding your answer.)

Worked solution: 1.4 / Q10

Step 1. Define the ticket counts.

Let a be adult tickets and c be child tickets. Then $a + c = 30$ and $12a + 8c = 300$.

Step 2. Solve for adult tickets.

Subtract $8(a + c) = 240$ from the revenue equation: $4a = 60$, so $a = 15$.

Step 3. Find adult-ticket spending.

$12a = 12(15) = 180$ dollars. Check: the other 15 tickets bring in \$120, for a total of \$300.

Final answer: 180

1.4 / Question 11

The solution to the given system of equations is (x, y) . What is the value of xy ?

$$5x + 14y = 45$$

$$10x + 7y = 27$$

Worked solution: 1.4 / Q11

Step 1. Eliminate x .

Double $5x + 14y = 45$ to get $10x + 28y = 90$. Subtract $10x + 7y = 27$: $21y = 63$.

Step 2. Find both variables.

$y = 3$. Then $5x + 14(3) = 45$ gives $5x = 3$, so $x = \frac{3}{5}$.

Step 3. Compute and check the product.

$xy = \frac{3}{5}(3) = \frac{9}{5} = 1.8$. Check: $10(\frac{3}{5}) + 7(3) = 6 + 21 = 27$.

Final answer: $\frac{9}{5}$ or 1.8

1.4 / Question 12

In the system of equations below, c is a constant. If the system has no solution, what is the value of c ?

$$y = \frac{1}{2}x + 8$$
$$y = cx + 10$$

Worked solution: 1.4 / Q12

Step 1. Identify the slopes and intercepts.

The first line has slope $\frac{1}{2}$ and intercept 8. The second has slope c and intercept 10.

Step 2. Require distinct parallel lines.

A system has no solution when the slopes match but the intercepts differ.

Step 3. Set the slopes equal.

$c = \frac{1}{2}$. Since $8 \neq 10$, the lines are distinct and parallel.

Final answer: $\frac{1}{2}$

1.4 / Question 13

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

$$y = 2x + 1$$

$$y = ax - 8$$

A. $-\frac{1}{2}$

B. 0

C. 1

D. 2

Worked solution: 1.4 / Q13

Step 1. Read the two slopes.

The slopes of $y = 2x + 1$ and $y = ax - 8$ are 2 and a .

Step 2. Apply the no-solution condition.

The slopes must be equal, so $a = 2$.

Step 3. Verify that the lines are distinct.

Their y -intercepts are 1 and -8 . Subtracting the equations with $a = 2$ would give $0 = 9$, an impossibility.

Final answer: D; 2

1.4 / Question 14

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

$$\begin{aligned}8x + 7y &= 9 \\ 24x + 21y &= 27\end{aligned}$$

A. $(r, -\frac{8r}{7} + \frac{9}{7})$

B. $(-\frac{8r}{7} + \frac{9}{7}, r)$

C. $(-\frac{8r}{7} + 9, \frac{9r}{7} + 27)$

D. $(\frac{r}{3} + 9, -\frac{r}{3} + 27)$

Worked solution: 1.4 / Q14

Step 1. Recognize equivalent equations.

$24x + 21y = 27$ is 3 times $8x + 7y = 9$, so both describe the same line.

Step 2. Choose a free coordinate.

Let $x = r$. Then $8r + 7y = 9$, giving $y = -\frac{8}{7}r + \frac{9}{7}$.

Step 3. Verify for every real r .

$8r + 7(-\frac{8}{7}r + \frac{9}{7}) = 9$. Multiplying by 3 also satisfies the second equation.

Final answer: A; $(r, -\frac{8r}{7} + \frac{9}{7})$

1.4 / Question 15

Store A sells raspberries for \$5.50 per pint and blackberries for \$3.00 per pint. Store B sells raspberries for \$6.50 per pint and blackberries for \$8.00 per pint. A certain purchase of raspberries and blackberries would cost \$37.00 at Store A or \$66.00 at Store B. How many pints of blackberries are in this purchase?

A. 4

B. 5

C. 8

D. 12

Worked solution: 1.4 / Q15

Step 1. Write the cost equations.

Let r and b be pints of raspberries and blackberries: $5.5r + 3b = 37$ and $6.5r + 8b = 66$.

Step 2. Eliminate r .

Double both equations: $11r + 6b = 74$ and $13r + 16b = 132$. Multiply these by 13 and 11, respectively, and subtract: $98b = 490$.

Step 3. Solve and check.

$b = 5$. Then $r = 4$; costs are $5.5(4) + 3(5) = 37$ and $6.5(4) + 8(5) = 66$ dollars.

Final answer: B; 5 pints

1.4 / Question 16

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

A. -9

B. 0

C. 9

D. 18

Worked solution: 1.4 / Q16

Step 1. Simplify the first equation.

$4x - 9y = 9y + 5$ gives $4x - 18y = 5$.

Step 2. Rewrite the second equation.

$hy = 2 + 4x$ gives $4x - hy = -2$.

Step 3. Match coefficients but not constants.

For no solution, set $h = 18$. The equations then require the same expression to equal both 5 and -2 , which is impossible.

Final answer: D; 18

1.4 / Question 17

One of the two equations in a linear system is $2x + 6y = 10$. The system has no solution. Which of the following could be the other equation in the system?

A. $x + 3y = 5$

B. $x + 3y = -20$

C. $6x - 2y = 0$

D. $6x + 2y = 10$

Worked solution: 1.4 / Q17

Step 1. Simplify the given equation.

Divide $2x + 6y = 10$ by 2: $x + 3y = 5$.

Step 2. Look for the same left side with a different constant.

Choice B is $x + 3y = -20$. Its line is parallel to and distinct from the given line.

Step 3. Verify the contradiction.

Subtracting the equations gives $0 = 25$. Choice A is the same line, while C and D have different slopes and intersect it.

Final answer: B; $x + 3y = -20$

1.4 / Question 18

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

$$\begin{aligned}2x + 3y &= 7 \\10x + 15y &= 35\end{aligned}$$

A. $\left(\frac{7}{5} + r, -\frac{7}{5} + 35\right)$

B. $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$

C. $\left(r, \frac{2r}{3} + \frac{7}{3}\right)$

D. $\left(r, -\frac{3r}{2} + \frac{7}{2}\right)$

Worked solution: 1.4 / Q18

Step 1. Recognize the same line twice.

$10x + 15y = 35$ is 5 times $2x + 3y = 7$.

Step 2. Let the y -coordinate be free.

Set $y = r$. Then $2x + 3r = 7$, so $x = -\frac{3}{2}r + \frac{7}{2}$.

Step 3. Check all real parameter values.

$2\left(-\frac{3}{2}r + \frac{7}{2}\right) + 3r = 7$ for every real r . The second equation follows by multiplying by 5.

Final answer: B; $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$

1.4 / Question 19

One of the equations in a system of two linear equations is given:

$$y = 6x + 18$$

The system has no solution. Which equation could be the second equation in the system?

A. $-6x + y = 18$

B. $-6x + y = 22$

C. $-12x + y = 36$

D. $-12x + y = 18$

Worked solution: 1.4 / Q19

Step 1. Identify the slope and intercept.

The given line $y = 6x + 18$ has slope 6 and intercept 18.

Step 2. Rewrite the candidate in choice B.

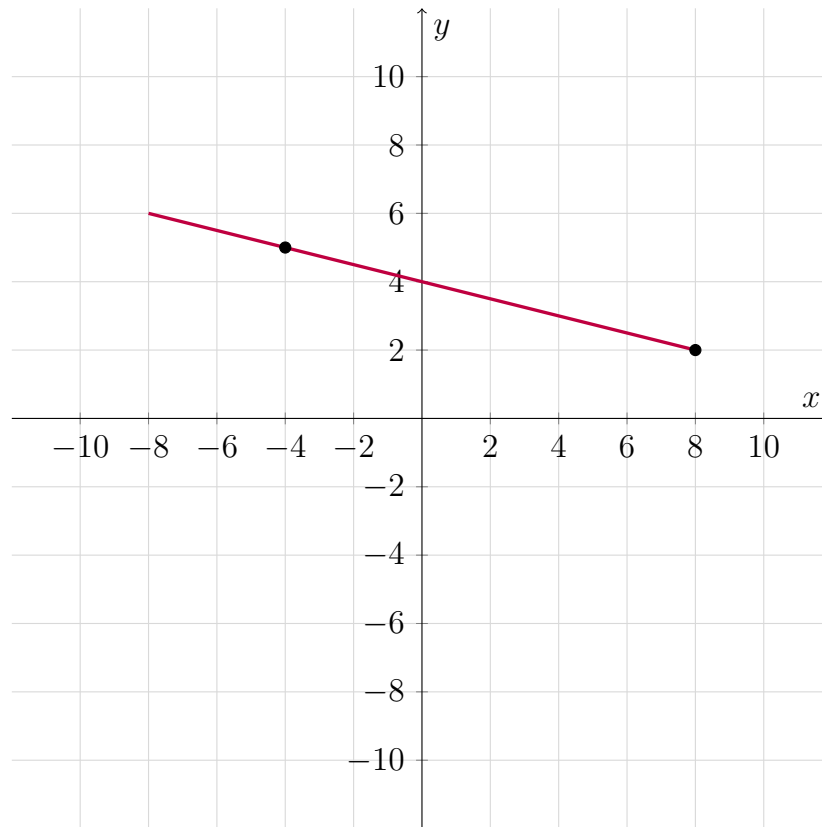
$-6x + y = 22$ becomes $y = 6x + 22$.

Step 3. Compare the two lines.

Equal slopes and different intercepts mean no solution. A is the same line; C and D have slope 12 and intersect the given line.

Final answer: B; $-6x + y = 22$

1.4 / Question 20



If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions (x, y) will the resulting system of three equations have?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Worked solution: 1.4 / Q20

Step 1. Find the line shown on the graph.

The marked points are $(-4, 5)$ and $(8, 2)$. Slope: $(2-5)/(8-(-4)) = -\frac{1}{4}$. The intercept is 4, so $y = -\frac{1}{4}x + 4$.

Step 2. Rewrite the new equation.

$x + 4y = -16$ gives $y = -\frac{1}{4}x - 4$.

Step 3. Determine whether all equations can hold.

The displayed line and the new line have the same slope but different intercepts. They have no common point, so the three-equation system has no solution.

Final answer: A; zero solutions

1.5 Linear Inequalities

1.5 / Question 1

Adam's school is a 20-minute walk or a 5-minute bus ride away from his house. The bus runs once every 30 minutes, and the number of minutes, w , that Adam waits for the bus varies between 0 and 30. Which of the following inequalities gives the values of w for which it would be faster for Adam to walk to school?

A. $w - 5 < 20$

B. $w - 5 > 20$

C. $w + 5 < 20$

D. $w + 5 > 20$

Worked solution: 1.5 / Q1

Step 1. Write each travel time.

Walking takes 20 minutes. Taking the bus takes $w + 5$ minutes, including waiting.

Step 2. Express “walking is faster.”

The walking time must be smaller: $20 < w + 5$, equivalently $w + 5 > 20$.

Step 3. Check the boundary.

This gives $w > 15$ within the given range $0 \leq w \leq 30$. At $w = 15$, the times are equal, so the inequality is strict.

Final answer: D; $w + 5 > 20$

1.5 / Question 2

A laundry service is buying detergent and fabric softener from its supplier. The supplier will deliver no more than 300 pounds in a shipment. Each container of detergent weighs 7.35 pounds, and each container of fabric softener weighs 6.2 pounds. The service wants to buy at least twice as many containers of detergent as containers of fabric softener. Let d represent the number of containers of detergent, and let s represent the number of containers of fabric softener, where d and s are nonnegative integers. Which of the following systems of inequalities best represents this situation?

A. $7.35d + 6.2s \leq 300$,
 $d \geq 2s$

B. $7.35d + 6.2s \leq 300$,
 $2d \geq s$

C. $14.7d + 6.2s \leq 300$,
 $d \geq 2s$

D. $14.7d + 6.2s \leq 300$,
 $2d \geq s$

Worked solution: 1.5 / Q2

Step 1. Write the shipment-weight limit.

Detergent weighs 7.35 d pounds and softener weighs 6.2 s pounds. “No more than 300” gives $7.35d + 6.2s \leq 300$.

Step 2. Write the container-count condition.

“At least twice as many detergent containers” means $d \geq 2s$.

Step 3. Combine both conditions.

Use $7.35d + 6.2s \leq 300$ and $d \geq 2s$, with d, s nonnegative integers. The factor 2 applies to the count comparison, not to the detergent’s weight coefficient.

Final answer: A

1.5 / Question 3

A local transit company sells a monthly pass for \$95 that allows an unlimited number of trips of any length. Tickets for individual trips cost \$1.50, \$2.50, or \$3.50, depending on the length of the trip. What is the minimum number of trips per month for which a monthly pass could cost less than purchasing individual tickets for trips?

Worked solution: 1.5 / Q3

Step 1. Use the largest individual-ticket price.

To make a pass cheaper with as few trips as possible, each trip can cost \$3.50. Thus n trips can cost at most $3.50n$.

Step 2. Require the pass to be strictly cheaper.

$95 < 3.50n$, so $n > 95/3.50 = 27\frac{1}{7}$.

Step 3. Choose and check the smallest integer.

The minimum is 28 trips. At 27 trips, even the highest cost is \$94.50; at 28 trips, it can be \$98, exceeding the \$95 pass price.

Final answer: 28 trips

1.5 / Question 4

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

$$y \leq x + 7$$

$$y \geq -2x - 1$$

A. $(-14, 0)$

B. $(0, -14)$

C. $(0, 14)$

D. $(14, 0)$

Worked solution: 1.5 / Q4

Step 1. Test choice D in the first inequality.

For $(14, 0)$, $0 \leq 14 + 7 = 21$, which is true.

Step 2. Test it in the second inequality.

$0 \geq -2(14) - 1 = -29$, also true.

Step 3. Check that the other choices fail.

A fails $0 \leq -7$; B fails $-14 \geq -1$; C fails $14 \leq 7$. Thus only D satisfies both conditions.

Final answer: D; $(14, 0)$

1.5 / Question 5

A salesperson's total earnings consist of a base salary of x dollars per year, plus commission earnings of 11% of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least 3 times and at most 4 times the base salary. Which of the following inequalities represents all possible values of total sales s , in dollars, the salesperson can make this year in order to meet that goal?

A. $2x \leq s \leq 3x$

B. $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$

C. $3x \leq s \leq 4x$

D. $\frac{3}{0.11}x \leq s \leq \frac{4}{0.11}x$

Worked solution: 1.5 / Q5

Step 1. Write total earnings.

Base salary plus commission gives $E = x + 0.11s$.

Step 2. Translate the earnings goal.

$3x \leq x + 0.11s \leq 4x$. Subtract x throughout: $2x \leq 0.11s \leq 3x$.

Step 3. Solve for sales.

Divide by the positive number 0.11: $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$.

Final answer: B; $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$

1.5 / Question 6

A number x is at most 2 less than 3 times the value of y . If the value of y is -4 , what is the greatest possible value of x ?

Worked solution: 1.5 / Q6

Step 1. Translate the wording.

“2 less than 3 times y ” is $3y - 2$. “At most” means $x \leq 3y - 2$.

Step 2. Substitute $y = -4$.

$$x \leq 3(-4) - 2 = -14.$$

Step 3. Choose the greatest allowable value.

Equality is permitted, so the greatest possible x is -14 .

Final answer: -14

1.5 / Question 7

A small business owner budgets \$2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays \$4.90 per candle to purchase small candles and \$11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

Worked solution: 1.5 / Q7

Step 1. Set up the minimum-cost purchase for a given number of large candles.

Let L and S be the numbers of large and small candles. Require $S + L \geq 200$ and $4.90S + 11.60L \leq 2200$. Buying 200 large candles already exceeds the budget, so feasible $L < 200$.

Step 2. Use the fewest small candles required.

For fixed L , choose $S = 200 - L$ to minimize cost. Then $4.90(200 - L) + 11.60L \leq 2200$, so $980 + 6.70L \leq 2200$ and $L \leq 182.089\dots$

Step 3. Check the greatest integer.

$L = 182$, $S = 18$ cost $11.60(182) + 4.90(18) = 2199.40$ dollars. The next choice, $L = 183$, $S = 17$, costs \$2206.10, over budget.

Final answer: 182 large candles

1.5 / Question 8

A psychologist set up an experiment to study the tendency of a person to select the first item when presented with a series of items. In the experiment, 300 people were presented with a set of five pictures arranged in random order. Each person was asked to choose the most appealing picture. Of the first 150 participants, 36 chose the first picture in the set. Among the remaining 150 participants, p people chose the first picture in the set. If more than 20% of all participants chose the first picture in the set, which of the following inequalities best describes the possible values of p ?

A. $p > 0.20(300 - 36)$, where $p \leq 150$

B. $p > 0.20(300 + 36)$, where $p \leq 150$

C. $p - 36 > 0.20(300)$, where $p \leq 150$

D. $p + 36 > 0.20(300)$, where $p \leq 150$

Worked solution: 1.5 / Q8

Step 1. Count all first-picture selections.

The first group contributes 36 people; the second contributes p . Total: $36 + p$.

Step 2. Translate “more than 20%.”

$$36 + p > 0.20(300).$$

Step 3. Include the second-group limit.

Because the second group has 150 people, $p \leq 150$. Equivalently, the possible integer values satisfy $24 < p \leq 150$.

Final answer: D; $p + 36 > 0.20(300)$, where $p \leq 150$

1.5 / Question 9

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 6 and 12, which inequality represents the possible lengths, x , of the third side of the triangle?

A. $x < 18$

B. $x > 18$

C. $6 < x < 18$

D. $x < 6$ or $x > 18$

Worked solution: 1.5 / Q9

Step 1. Use the upper-bound triangle inequality.

The known sides must sum to more than the third: $6 + 12 > x$, so $x < 18$.

Step 2. Use the lower-bound triangle inequality.

$6 + x > 12$ gives $x > 6$. The remaining inequality, $12 + x > 6$, is automatically true when $x > 6$.

Step 3. Combine the restrictions.

$6 < x < 18$. The endpoints would produce a degenerate triangle, so neither is included.

Final answer: C; $6 < x < 18$

1.5 / Question 10

The formula above is Ohm's law for an electric circuit with current I , in amperes, potential difference V , in volts, and resistance R , in ohms.

$$I = \frac{V}{R}$$

A circuit has a resistance of 500 ohms, and its potential difference will be generated by n six-volt batteries that produce a total potential difference of $6n$ volts. If the circuit is to have a current of no more than 0.25 ampere, what is the greatest number, n , of six-volt batteries that can be used?

Worked solution: 1.5 / Q10

Step 1. Substitute voltage and resistance.

Ohm's law gives $I = V/R = 6n/500$.

Step 2. Apply the current limit.

$6n/500 \leq 0.25$ gives $6n \leq 125$, hence $n \leq 125/6 = 20\frac{5}{6}$.

Step 3. Choose and check an integer.

At most 20 batteries can be used. Their current is $120/500 = 0.24$ ampere; 21 batteries produce $126/500 = 0.252$ ampere, too much.

Final answer: 20 batteries

1.5 / Question 11

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

$$y < 6x + 2$$

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

Worked solution: 1.5 / Q11

Step 1. Calculate the strict upper bounds.

For $x = 3, 5, 7$, the values of $6x + 2$ are 20, 32, and 44.

Step 2. Test every row of choice C.

Its values satisfy $16 < 20$, $28 < 32$, and $40 < 44$.

Step 3. Eliminate the other tables.

A has equality in every row, which fails the strict inequality. B fails at $x = 5$ because $36 \not< 32$. D already fails at $x = 3$ because $24 \not< 20$.

Final answer: C

1.5 / Question 12

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

$$y > 13x - 18$$

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

Worked solution: 1.5 / Q12

Step 1. Calculate the strict lower bounds.

At $x = 3, 5, 8$, the expression $13x - 18$ equals 21, 47, and 86.

Step 2. Test all rows in choice D.

$26 > 21$, $52 > 47$, and $91 > 86$, so all rows satisfy the inequality.

Step 3. Eliminate the other tables.

A uses equality and fails. B fails at $x = 5$ because $42 \not> 47$; C fails at $x = 3$ because $16 \not> 21$.

Final answer: D

1.5 / Question 13

A shipping service restricts the dimensions of the boxes it will ship for a certain type of service. The restriction states that for boxes shaped like rectangular prisms, the sum of the perimeter of the base of the box and the height of the box cannot exceed 130 inches. The perimeter of the base is determined using the width and length of the box. If a box has a height of 60 inches and its length is 2.5 times the width, which inequality shows the allowable width x , in inches, of the box?

A. $0 < x \leq 10$

B. $0 < x \leq 11\frac{2}{3}$

C. $0 < x \leq 17\frac{1}{2}$

D. $0 < x \leq 20$

Worked solution: 1.5 / Q13

Step 1. Express the base perimeter.

Width is x and length is $2.5x$. Thus the perimeter is $2x + 2(2.5x) = 7x$.

Step 2. Apply the shipping restriction.

Perimeter plus height cannot exceed 130: $7x + 60 \leq 130$.

Step 3. Solve and include positive dimensions.

$7x \leq 70$ gives $x \leq 10$. Since a box must have positive width, $0 < x \leq 10$.

Final answer: A; $0 < x \leq 10$

1.5 / Question 14

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is \$14,000, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

A. \$148.15

B. \$161.53

C. \$172.84

D. \$184.94

Worked solution: 1.5 / Q14

Step 1. Define the pre-tax unit price.

Let p be the price per chair before sales tax. The subtotal for 81 chairs is $81p$.

Step 2. Include the tax and budget.

A 7% sales tax multiplies the subtotal by 1.07: $1.07(81p) \leq 14,000$.

Step 3. Find the maximum.

$p \leq \frac{14,000}{86.67} \approx 161.53225$. The closest choice is \$161.53. Check: $86.67(161.53) = 13,999.8051$, within budget.

Final answer: B; \$161.53

1.5 / Question 15

Ken is working this summer as part of a crew on a farm. He earned \$8 per hour for the first 10 hours he worked this week. Because of his performance, his crew leader raised his salary to \$10 per hour for the rest of the week. Ken saves 90% of his earnings from each week. What is the least number of hours he must work the rest of the week to save at least \$270 for the week?

A. 38

B. 33

C. 22

D. 16

Worked solution: 1.5 / Q15

Step 1. Calculate earnings from the first 10 hours.

$8(10) = 80$ dollars. Let h be the hours worked during the rest of the week; total earnings are $80 + 10h$.

Step 2. Apply the savings goal.

Ken saves 90%: $0.90(80 + 10h) \geq 270$.

Step 3. Solve and check.

Divide by 0.90: $80 + 10h \geq 300$, so $h \geq 22$. At 22 additional hours, savings are $0.90(300) = 270$ dollars.

Final answer: C; 22 additional hours

1.5 / Question 16

$$\begin{aligned}y &> 2x - 1 \\ 2x &> 5\end{aligned}$$

Which of the following consists of the y -coordinates of all the points that satisfy the system of inequalities above?

A. $y > 6$

B. $y > 4$

C. $y > \frac{5}{2}$

D. $y > \frac{7}{2}$

Worked solution: 1.5 / Q16

Step 1. Find the necessary lower bound.

From $2x > 5$, $x > \frac{5}{2}$. Therefore $2x - 1 > 4$. Since $y > 2x - 1$, every solution has $y > 4$.

Step 2. Show that every $y > 4$ is possible.

For a fixed $y > 4$, the first inequality requires $x < \frac{y+1}{2}$. Together, $\frac{5}{2} < x < \frac{y+1}{2}$. This interval is nonempty because $y > 4$.

Step 3. State the full set of y -coordinates.

All and only values $y > 4$ occur. The boundary $y = 4$ is excluded by the strict inequalities.

Final answer: B; $y > 4$

Quick Answer Reference

1.1 Linear Equations in One Variable

Question	Answer
1	D; 12 hours
2	D; $\frac{4}{25}m = 5$
3	C; infinitely many solutions
4	$\frac{16}{17}$
5	A; $(100(1 - 0.2) - 65)u = 6840$
6	$-\frac{14}{15}$
7	-22
8	B; approximately \$291,700
9	B; $6n + 6 = 30$
10	B; $p = \frac{1}{7}$
11	B; between -2 and 2
12	D; I and III only
13	C; $2x + 20$

1.2 Linear Functions

Question	Answer
1	A; $f(x) = 60x + 100$
2	A; 16,250 units
3	D; $y = 135x + 135$
4	A; 16.38 degrees Fahrenheit
5	-12
6	D; the average price per gallon on December 1, 2014
7	D; 192
8	C; $f(t) = -\frac{21}{130}t + 4$
9	A; $P = \frac{w}{110}$
10	A; 3.78 degrees Fahrenheit
11	B; $(-4, 0)$
12	B; $f = 20 - \frac{4}{9}(p + c)$
13	D; length increases by 2 cm for each additional newton
14	B; -4
15	A; $f(n) = 14n + 175$

1.3 Linear Equations in Two Variables

Question	Answer
1	33
2	A; -2
3	D; $(-\frac{17}{6}, 0)$
4	5 hours
5	$\frac{1}{2}$
6	74
7	24
8	A; $f(x) = -\frac{1}{4}x - 12$
9	C; the total cups of broth in the large jars
10	C
11	\$1.50
12	D; $\frac{7}{2}$
13	A; 0
14	$\frac{59}{9}$
15	30
16	B
17	40 points
18	B; 160
19	$\frac{5}{4}$
20	C; $\frac{24}{7}$
21	D; $x - y = -3$
22	1.5 liters

1.4 Systems of Two Linear Equations

Question	Answer
1	1.5
2	6
3	8
4	D; 160 miles
5	D; $\frac{3}{2}$
6	35
7	B; -5
8	-34
9	80
10	180
11	$\frac{9}{5}$ or 1.8
12	$\frac{1}{2}$
13	D; 2
14	A; $\left(r, -\frac{8r}{7} + \frac{9}{7}\right)$
15	B; 5 pints
16	D; 18
17	B; $x + 3y = -20$
18	B; $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$
19	B; $-6x + y = 22$
20	A; zero solutions

1.5 Linear Inequalities

Question	Answer
1	D; $w + 5 > 20$
2	A
3	28 trips
4	D; $(14, 0)$
5	B; $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$
6	-14
7	182 large candles
8	D; $p + 36 > 0.20(300)$, where $p \leq 150$
9	C; $6 < x < 18$
10	20 batteries
11	C
12	D
13	A; $0 < x \leq 10$
14	B; \$161.53
15	C; 22 additional hours
16	B; $y > 4$